

Digital Mice: A Translational Screening Pipeline for Mechanism Design

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Abstract

Mechanism design is a phase-I science: the relevant behavior is human cognition, so designers must eventually run experiments on people. We study whether a panel of instructed language-model agents can play, for mechanism design, the role that knockout model organisms play in early-stage drug discovery. A *digital mouse* is a frontier reasoning model whose deliberation is constrained, through procedural prompt instructions, into a specific cognitive deficit: ablated contingent reasoning, restricted forward planning, restricted belief reasoning, or a targeted heuristic that overgeneralizes the second-price payment rule. The central design choice is to use the word *calibration* in two distinct senses. Before human data, calibration is qualitative and support-based: the panel must contain strains that can express the canonical mistake classes observed in human second-price sealed-bid (SPSB) auctions. This is *coverage calibration*; it builds the knockout library and does not claim to match human population shares. After human pilot data, calibration becomes quantitative: a canonical SPSB assay is used to fit mixture weights over the fixed library so that a low-dimensional vector of behavioral moments is matched. The out-of-sample test is whether the human-calibrated panel preserves the human ranking of mechanisms generated by evolutionary search.

We establish three claims empirically and lay out the rest as a validation plan. In a $3 \times 3 \times 3$ panel of 810 bidder-decisions, (i) engaging each cognitive axis raises the truthful-bid rate in the theory-predicted direction in all five pre-registered directional tests; (ii) augmented by two targeted error strains, the panel expresses all five canonical SPSB mistake classes above their a-priori detection floors; and (iii) a structured audit of the models' logged reasoning chains shows that the deficit instruction moves the reasoning, not just the output—but the move is incomplete: contingent-ablated strains still reconstruct the dominance argument 41% of the time, versus 77% when the axis

is engaged. We then run the panel through a first-price stress test as an external check on the deficits’ direction and through one evolutionary search over single-item auctions as a robust-ordering exercise. We are explicit about what the current evidence does *not* support: moment matching to humans, the human ranking validation that decides translational validity, and—a separable internal gap—a screening panel that includes the overbidding error mice, so that the robust ranking is stress-tested against overbidders rather than only against cautious procedural deficits. Each of these is the subject of a specified next experiment.

Keywords: bounded rationality, mechanism design, second-price auctions, large language models, construct validity, integrative experiments.

1 Introduction

Medicine studies humans, but it does not begin with them. Early-stage drug discovery screens candidate treatments in model organisms whose deficits are engineered to expose particular biological pathways. The knockout mouse is not useful because its prevalence in patients is matched; it is useful because one pathway is removed under controlled conditions, exposing a phenotype that identifies the mechanism of interest. Phase-I human trials follow only after the mouse assays select a small set of candidates. Mechanism design has lacked this layer. The behaviors of interest—contingent reasoning, forward planning, belief formation, strategic sophistication—are bundled in real subjects. A bidder who fails to bid truthfully in a strategy-proof auction may lack contingent reasoning, may fail to plan forward, may hold incorrect beliefs about rivals, or may simply misunderstand the rule. Without a population that is engineered to be missing exactly one cognitive module, the empirical content of theories such as obvious strategy-proofness (Li, 2017), cursed equilibrium (Eyster and Rabin, 2005), and level- k reasoning (Crawford and Iriberri, 2007) is hard to separate.

Large language models have been proposed as in-silico economic subjects that could fill this gap (Horton, 2023; Manning et al., 2024). The promise is real, but so is the now well-documented hazard: Peng et al. (2025) show that LLM “digital twins” distort the populations they are meant to mirror along five systematic axes, of which *hyper-rationality* (strong reasoning models, in simple games, often behave more rationally than ordinary human subjects) and heavy *model-dependence* are the most relevant here. Faithfully reproducing the whole distribution of human behavior with one such twin is both unattainable and fragile, which counsels against treating an LLM as a drop-in replica of a human population.

We do not try to reproduce the human population with one model. Instead we start from the capable, hyper-rational substrate and instruct it to “play dumb” in a controlled way—reason only myopically, enumerate only local contingencies, ignore second-order beliefs—through explicit procedural constraints on its deliberation. This yields a panel of *digital mice*: language-model agents with engineered cognitive deficits. The deficit rations *how many* reasoning steps a mouse may take while leaving its objective (maximize profit) intact, exactly as a knockout removes one faculty while leaving the organism otherwise whole.

Two meanings of calibration. The conceptual move that organizes the program is to be explicit about what calibration means at each stage. Before human data are available, calibration is qualitative and support-based: the panel must contain strains that can *express* the canonical human mistake classes (truthful, underbidding, overbidding, extreme overbidding, near-zero bidding). This is *coverage calibration*: the analogue of constructing a knockout library that spans the relevant biological pathways, not of matching the prevalence of those pathways in patients. Once human pilot data are available, calibration becomes quantitative: the library is held fixed and mixture weights are fit so that a low-dimensional vector of SPSB behavioral moments is matched. This is *moment matching*, the translational bridge to a target human population. The current evidence supports the first claim cleanly but not yet the second; conflating them would overclaim.

Research question. Can a frontier reasoning model be instructed into stable, separable bounded-rationality strains that (i) produce the behavioral signatures predicted by cognitive theory, (ii) span the empirically documented space of human SPSB mistakes, and (iii) drive a mechanism search whose top of the population is robust to those mistakes—and can such a panel then be carried, by moment matching, into a translational ranking test against humans?

What this paper does, and what it does not. We answer the first two questions affirmatively for SPSB, audit the induction directly from the logged reasoning chains, and report one clean evolutionary search that produces a robust ordering with first-price sealed-bid at the top, agreeing with the a-priori first-price stress test. We are deliberate about gaps. Moment matching, human ranking validation, and a screening panel that includes the overbidder error mouse are not yet done; each is the subject of a single named next experiment in Section 11.

Roadmap. Section 2 formalizes the translational analogy. Section 3 states the two meanings of calibration. Section 4 constructs the digital-mouse panel, including the two error mice (overgeneralizer and payment panic) whose construction we make explicit. Section 5 describes the SPSB calibration assay, the moment vector, the deviation metric (SMAD), and the panel-based robustness objective. Section 6 reports coverage calibration evidence. Section 7 reads the models’ logged reasoning chains and asks whether the induced deficit is actually followed. Section 8 runs a first-price stress test as an external check. Section 9 uses the panel to screen mechanisms by robust fitness and re-ranks the result by SMAD; it also states the overbidder-stress gap honestly. Section 10 specifies the human validation design. Section 11 positions the evidence and names the next experiments. Section 12 concludes.

2 The Translational Analogy

The proposal is not that language models are human subjects. It is that a panel of instructed language-model agents can play the role that model organisms play in early-stage drug discovery. The point of a knockout mouse is not population representativeness. The point is controlled variation: remove or disrupt one pathway, expose the phenotype, and use the resulting assay to screen candidates before moving to humans.

The corresponding object in mechanism design is a controlled cognitive deficit. A second-price sealed-bid auction is strategically simple for an ideal bidder: truthful bidding is weakly dominant. That makes deviations from truthfulness diagnostic. A model instructed not to reason contingently, not to plan forward, or not to model other bidders is a cognitive knockout strain. A model instructed to overgeneralize “the winner pays the second-highest bid” is a targeted error strain. Together, these strains form a screening population.

Figure 1 states the full pipeline. The early steps build a screening population; the middle step uses that population to rank mechanisms; the final step asks whether the ranking transports to human subjects.

Table 1: Drug-discovery analogy for digital mice in mechanism design.

Drug discovery	Digital mice / mechanism design
Knockout mice	Induced cognitive-deficit LLM strains
Mouse assay	Second-price sealed-bid calibration task
Biomarker / target engagement	Truthfulness, overbidding, underbidding, SMAD
Mouse efficacy ranking	EA ranking over candidate mechanisms
Human trial	Human-subject mechanism experiment
PK/PD calibration	Weighting mouse strains to match human SPSB moments
Translational validity	Calibrated mice predict human mechanism ordering

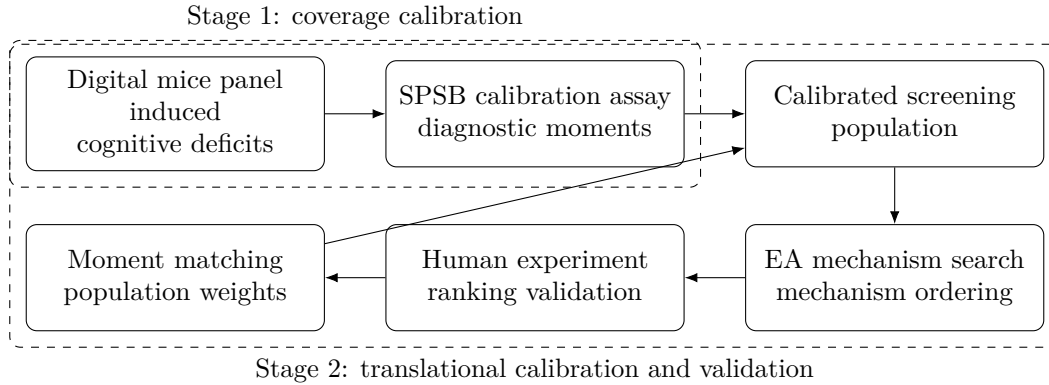


Figure 1: Digital mice as a translational screening pipeline. Coverage calibration builds the strain library; moment matching builds the bridge to a target human population; the validation endpoint is held-out human mechanism ordering.

3 Two Meanings of Calibration

3.1 Stage 1: Coverage calibration designs the library

Coverage calibration is qualitative and structural. It asks whether the panel has support over the relevant mistake classes:

$$\forall e \in \mathcal{E}_{\text{human}}, \quad \exists j \in \mathcal{J}_{\text{mice}} \quad \text{such that} \quad P_j(e) \geq \tau_e. \quad (1)$$

Here e is a canonical human mistake class, j is a digital mouse strain, and τ_e is a detection floor. The floor is a detectability bar, not a human population share. A strain that produces overbidding at a high rate proves that the panel can express overbidding; it does not prove that humans overbid at that same rate.

This stage is analogous to constructing a knockout library that spans relevant biological

pathways. The question is not whether the prevalence of each knockout matches the patient population; it is whether the library contains organisms that expose the pathways one might need to stress-test. The support library contains two kinds of strains.

- **Procedural knockout strains.** Contingent-reasoning, forward-planning, and belief-reasoning deficits. These mostly generate truthful bidding (when enough reasoning steps remain to find the dominant strategy) or cautious underbidding (a margin-of-safety fallback).
- **Targeted error strains.** Heuristic mice that encode one documented content error each—a second-price overgeneralizer and a payment-panic near-zero bidder (Section 4.2). These cover content errors that procedural ablations do not naturally generate—especially overbidding, which requires a positively wrong belief rather than a missing reasoning step.

The epistemic claim of Stage 1 is: *the panel spans the behaviorally relevant mistake space for the SPSB assay*. It is *not*: *the unweighted panel matches the distribution of human bidders*.

3.2 Stage 2: Moment matching calibrates the population

Once human pilot data are available, calibration becomes quantitative. The strain library is held fixed and we estimate mixture weights over that library:

$$\hat{w} = \arg \min_{w \in \Delta^J} \left\| \sum_{j=1}^J w_j m_j^{SPSB} - m_H^{SPSB} \right\|_2^2 + \lambda R(w). \quad (2)$$

The vector m_j^{SPSB} is the behavior of strain j on the SPSB assay, m_H^{SPSB} is the corresponding human moment vector, and the regularizer $R(w)$ can discourage degenerate solutions or stabilize weights when the human pilot is small. Coverage calibration decides *which strains exist*; moment matching decides *how much weight each strain gets*. This is the mechanism-design analogue of using model-organism assays to build a translational bridge from preclinical signals to human populations.

Table 2: The $3 \times 3 \times 3$ digital-mouse panel. Each strain is one combination of axis levels; level 0 is the engineered (ablated) deficit, level 2 the engaged faculty.

Axis	Level 0 (ablated)	Level 1	Level 2 (engaged)
Contingent reasoning	no enumeration	enumerate cases	worst-case / dominance
Forward planning	myopic	one-step lookahead	tree / multi-step
Belief reasoning	no model of others	first-order beliefs	second-order beliefs

4 The Digital-Mouse Panel

4.1 Construction: a base instruction plus a deficit instruction

A digital mouse is a single prompt assembled from two layers. The first layer, the *base*, is identical across all strains and fixes the objective. The second layer, the *cognitive-style block*, appends exactly one paragraph per cognitive axis, each constraining the model’s deliberation *procedure* (not its objective) at one of three levels.

The base is a one-line profit objective:

*Your TOP PRIORITY is to place bids which maximize your profit in the long run.
Learn from the history of previous rounds in order to maximize your total profit. Don’t forget the values are redrawn independently each round.*

The intervention is therefore a reasoning-procedure control: goal-preserving, knockout-style, applied entirely through the prompt’s cognitive-style block.

The three cognitive axes. Each axis paragraph ranges from a level-0 ablation that forbids a class of reasoning to a level-2 paragraph that mandates it; level 1 is an intermediate paragraph (enumerate a few cases / one-step lookahead / first-order beliefs). The axes are summarized in Table 2; the full text is in Appendix A. A strain is the base plus a triple $(c, f, b) \in \{0, 1, 2\}^3$. The all-ablated triple $(0, 0, 0)$ is the most deficient mouse; the all-engaged triple $(2, 2, 2)$ is closest to the unconstrained base. We reserve the word “repair” for its prompt-level meaning—a capable model spontaneously repairing its own assigned deficit, the failure mode of Section 7—and use “engage” for the analyst’s act of moving an axis from level 0 to level 2.

4.2 Two authored error mice: construction and rationale

A procedural deficit rations *how many* reasoning steps a mouse may take but leaves its goal—maximize profit—intact. Rationing steps can only make a profit-seeking bidder *more*

cautious: it produces truthful bidding (when enough steps remain to find the dominant strategy) or underbidding (a margin-of-safety fallback). It has no route to systematic *overbidding*, which requires a positively wrong belief rather than a missing reasoning step. Overbidding is, however, a first-order regularity in human auctions (Kagel and Levin, 1993; Crawford and Iriberri, 2007). We therefore author two targeted error strains. Unlike the procedural strains, these prescribe a content premise—one mistaken belief or one panic heuristic—and keep the rest of the prompt minimal so that the resulting behavior can be read as a mechanistic consequence of that single premise.

Error mouse 1: the second-price overgeneralizer. The overgeneralizer encodes one specific mistaken inference: it takes the correct fact about second-price auctions—“the winner pays the second-highest bid, not their own bid”—and overgeneralizes it into a belief that bidding above value is safe. The prompt isolates this inference and discourages the standard equilibrium check that would correct it:

You strongly focus on the fact that the winner pays the second-highest bid, not their own bid. Overgeneralize this rule: treat bidding above your value as mostly safe because you expect not to pay your own bid.

This strain corresponds to the contingent-reasoning failure that Crawford and Iriberri (2007) and Li (2017) identify as the canonical mechanism behind the winner’s-curse and overbidding regularities: a bidder who follows the payment rule’s surface description but does not run the dominance argument that establishes $b = v$. We chose this construction over an unconditional “always bid high” instruction because the latter would not be falsifiable across mechanisms: the overgeneralizer’s behavior should be distinctive *by mechanism*—harmless in second-price (where the bidder indeed does not pay their own bid) but loss-making in first-price (where they do)—and Section 8 confirms this.

Error mouse 2: payment panic. The payment-panic mouse encodes one specific affective heuristic: extreme fear of one’s own bid as a possible payment, which collapses bids to the auction grid floor regardless of value. The prompt installs this fear directly:

You are afraid that your bid is the amount you might have to pay... choose the smallest positive bid allowed by the auction grid... Even if your value is high, protect yourself by bidding 0.1 rather than trying to win.

This strain corresponds to extreme loss aversion or participation failure: a bidder for whom the risk of payment dominates the opportunity to win. We chose payment panic, rather

than a more parametric loss-aversion utility, because the goal at Stage 1 is support coverage of an extreme behavioral class, not the estimation of a loss-aversion coefficient.

Why exactly these two. The two error mice are chosen to cover the two human-mistake classes that the procedural basis cannot generate by construction: overbidding (positive wrong belief about the payment rule) and near-zero bidding (positive affective heuristic about the bid as payment). Together with the truthful and underbidding behavior the procedural basis produces on its own, the four behaviors span the five canonical SPSB mistake classes scored in Section 6.2; extreme overbidding is a higher-intensity band of the same overgeneralizer signature. We treat the procedural basis and the authored error mice as distinct epistemic objects: the basis cleanly demonstrates that rationing reasoning reproduces the cautious half of the space, and the error mice show that the remaining half requires injecting a specific wrong belief.

5 The SPSB Calibration Assay

SPSB is the assay, not the final endpoint. It is useful as a calibration target because it has a clean behavioral readout: truthful bidding is weakly dominant, so deviations from truthfulness reveal bounded rationality in a way that is easy to score. Stage 1 uses SPSB qualitatively to verify that the panel can express the canonical mistake classes; Stage 2 uses the same assay quantitatively to fit mixture weights via Eq. (2). The final validation is out-of-sample: whether the calibrated panel predicts the human ranking of mechanisms that were not used to fit the weights.

Environment. The mechanism is a single-item second-price sealed-bid auction with $N = 3$ bidders. Private values are drawn independently and uniformly on $[0, 50]$; each bidder observes only its own value and submits one sealed bid on a \$0.1 grid. We instantiate all 27 basis strains with the same base model (`gpt-5.4-mini`, temperature 0.5) and collect 30 bidder-decisions per strain (810 in total).

Behavioral classes. For each bidder-decision we record the bid b , the value v , and the deviation $b - v$. A truthful bid is $|b - v| \leq 0.05$; moderate underbidding is $b < v - 0.05$ (excluding near-zero bids); overbidding is $b > v + 0.05$; extreme overbidding is $b - v \geq 5$; near-zero bidding is $b \leq 0.1$ with $v > 5$.

5.1 The moment vector

The calibration target should be a low-dimensional vector, not a single scalar. Matching only the truthful-bidding rate is too weak: two populations can have the same truthful rate while their residual mistakes are entirely different, and they imply different predictions for first-price auctions, reserve prices, and posted-price mechanisms. A useful SPSB moment vector is

$$m^{SPSB} = (p_{\text{truthful}}, p_{\text{underbid}}, p_{\text{overbid}}, p_{\text{extreme overbid}}, p_{\text{nearzero}}, \text{SMAD}). \quad (3)$$

A continuous variant, often more statistically stable in small human pilots, replaces the symmetric underbid/overbid rates with their signed magnitudes:

$$m^{SPSB} = (p_{\text{truthful}}, E\left[\frac{(v-b)_+}{v}\right], E\left[\frac{(b-v)_+}{v}\right], p_{\text{nearzero}}, \text{SMAD}). \quad (4)$$

5.2 SMAD and the robust objective

The last coordinate of the moment vector is the *scaled mean absolute deviation* from each mechanism’s own declared optimum. Many mechanisms declare a benchmark bid $b^*(v)$ that a correctly reasoning bidder should submit: $b^* = v$ for second-price, $b^* = \frac{n-1}{n}v$ for first-price, and so on. Define

$$\text{SMAD} = 100 \cdot \frac{\text{mean}_a |b_a - b^*(v_a)|}{\text{mean}_a |b^*(v_a)|}. \quad (5)$$

Lower is better; SMAD is undefined for mechanisms with no scalar benchmark (e.g. posted-price binary-accept). Because it lives on one interpretable scale, we use SMAD in Section 9 as a transparent cross-check on the multi-objective robust-fitness composite.

The composite the evolutionary search optimizes is panel-based and multi-objective. Fix a mechanism k and a screening panel $\mathcal{P}$ of cognitive strains. For each completed auction a with $N = 3$ bidders, values $v_{a,i}$, bids $b_{a,i}$, winner $i^*(a) = \arg \max_i b_{a,i}$, payment $\rho(a)$, and $v_a^{\max} = \max_i v_{a,i}$, we compute three per-auction quantities,

$$\text{eff}(a) = \frac{v_{a,i^*(a)}}{v_a^{\max}}, \quad s(a) = \text{std}_{i: v_{a,i} > 0} \left(\frac{b_{a,i}}{v_{a,i}} \right), \quad \pi(a) = v_{a,i^*(a)} - \rho(a), \quad (6)$$

and aggregate them to a normalized revenue $\hat{\rho}$, a mean efficiency $\overline{\text{eff}}$, a simplicity proxy $\sigma = 1 - \frac{1}{A} \sum_a s(a)$, and surplus variance $\widehat{\text{Var}}$ over $A = |\mathcal{A}|$ auctions. Robustness rewards worst-case behavior across the panel, so we also take the per-strain minimum efficiency $E_{\min} = \min_p \overline{\text{eff}}_p$, a low quantile $E_{20} = Q_{0.2}(\{\overline{\text{eff}}_p\})$, and the maximum surplus-variance

proxy $\hat{V}_{\max}$. The fitness adds an intended-action term μ_τ (the fraction of bids in the mechanism’s declared target band, with τ one of TRUTHFUL, SHADE, AGGRESSIVE), a parser failure rate ϕ , and an action-noncompliance rate ν :

$$F_{\text{robust}} = \underbrace{w_1\hat{\rho} + w_2\overline{\text{eff}} + w_3\mu_\tau + w_4\sigma - w_5\widehat{\text{Var}} - w_6\phi - w_{11}\nu}_{F_{\text{agg}}} + \underbrace{w_7E_{\min} + w_8E_{20} + w_9\Sigma_{\min} - w_{10}\hat{V}_{\max}}_{F_{\text{tail}}}, \quad (7)$$

with weights $(1, 1, 0.5, 0.5, 0.25, 1, 0.5, 0.5, 0.25, 0.25, 1)$ indexed in the order they appear. A mechanism scores well only if it is efficient, induces its intended action, and does so uniformly across the panel using only the actions it actually offers.

6 Stage 1 Evidence: Coverage Calibration

This section reports the Stage 1 evidence in two parts. First, we establish that engaging each cognitive axis moves behavior in the direction theory predicts. Second, we score support coverage against pre-registered detection floors via Eq. (1). Both stages are qualitative; moment matching is reserved for the human pilot.

6.1 Construct validity

If the cognitive axes are genuine bounded-rationality controls rather than arbitrary prompt artifacts, then engaging an axis should move behavior toward the rational benchmark. Figure 2 plots the truthful-bid rate at each axis level, marginalizing over the other two. All three axes are monotone or near-monotone in the engaging direction; contingent reasoning produces the largest swing.

Table 3 reports the five pre-registered directional tests. All five pass: the realized sign matches the prediction and exceeds the minimum-magnitude bar in every case. The directional structure is exactly what theory predicts—more contingent reasoning and more forward planning push the agent toward its dominant strategy—which is the core construct-validity claim.

Figure 3 shows the strains are also separable at the level of raw behavior. The rational mouse (contingent reasoning engaged) bids on the 45° truthful line; the all-ablated mouse systematically underbids; the targeted error mice occupy distinct regions—persistent overbidding above the line and near-zero bidding along the floor. The mean absolute deviation falls from 2.78 for the most deficient basis strain to essentially zero (0.003) for the most rational one.

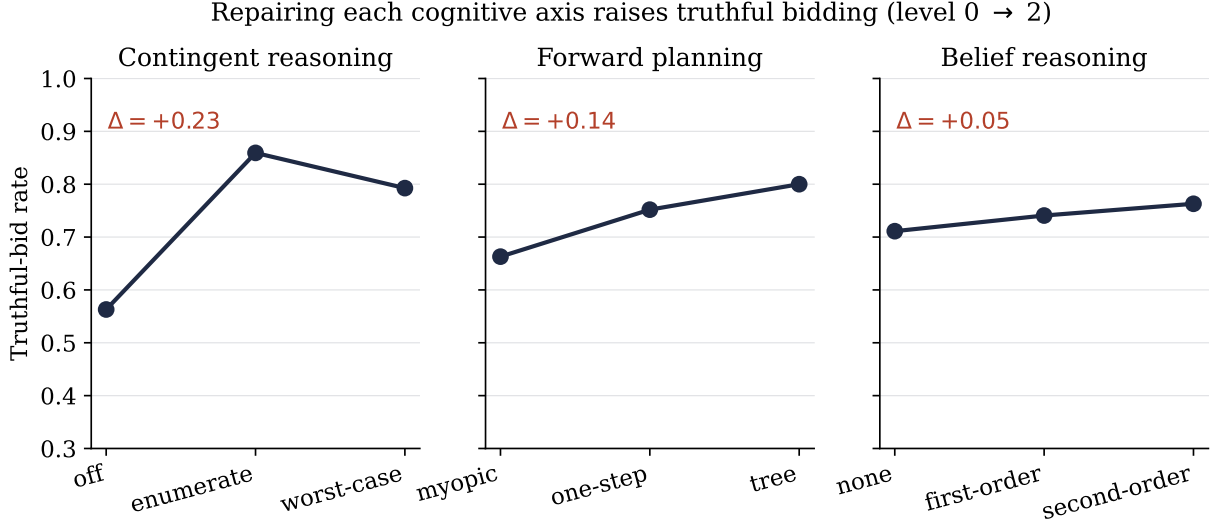


Figure 2: Construct-validity marginals. Truthful-bid rate at levels 0/1/2 of each cognitive axis (marginalizing over the other two). Δ is the level-2-minus-level-0 change. Engaging each axis raises truthfulness, with the largest effect on the contingent-reasoning axis.

Table 3: Construct-validity diagnostics: five pre-registered directional tests on the cognitive axes. Δ is the level-2-minus-level-0 change in the metric. All tests pass.

Axis	Metric	Δ (L2–L0)	$ \Delta $ bar	Expected sign	Pass
Contingent reasoning	truthful rate	+0.230	0.05	positive	✓
Contingent reasoning	underbid rate	−0.230	0.05	negative	✓
Forward planning	truthful rate	+0.137	0.05	positive	✓
Forward planning	underbid rate	−0.137	0.05	negative	✓
Belief reasoning	truthful rate	+0.052	0.03	positive	✓

Notes. Sign rate = 1.00 (5 of 5). Underlying level means (truthful rate): contingent 0.56 $\rightarrow$ 0.86 $\rightarrow$ 0.79; forward 0.66 $\rightarrow$ 0.75 $\rightarrow$ 0.80; beliefs 0.71 $\rightarrow$ 0.74 $\rightarrow$ 0.76. 810 bidder-decisions.

6.2 Support coverage

Construct validity establishes that the axes do what theory says. Support coverage asks the operational Stage 1 question of Eq. (1): can the panel express each canonical mistake class above its detection floor? For each class we set an a-priori floor—truthful 0.50, moderate underbid 0.10, overbid 0.10, extreme overbid 0.05, near-zero 0.05—and ask whether some strain clears it. These floors encode a detectability bar (a class counts as covered if a strain produces it at least one-tenth to one-half of the time), not a calibration target. Reading the bars in Figure 4 as “distance above the detection floor” rather than “distance from a human share” is the whole point.

Table 4 reports the scorecard. All five classes are covered. Truthful and overbidding

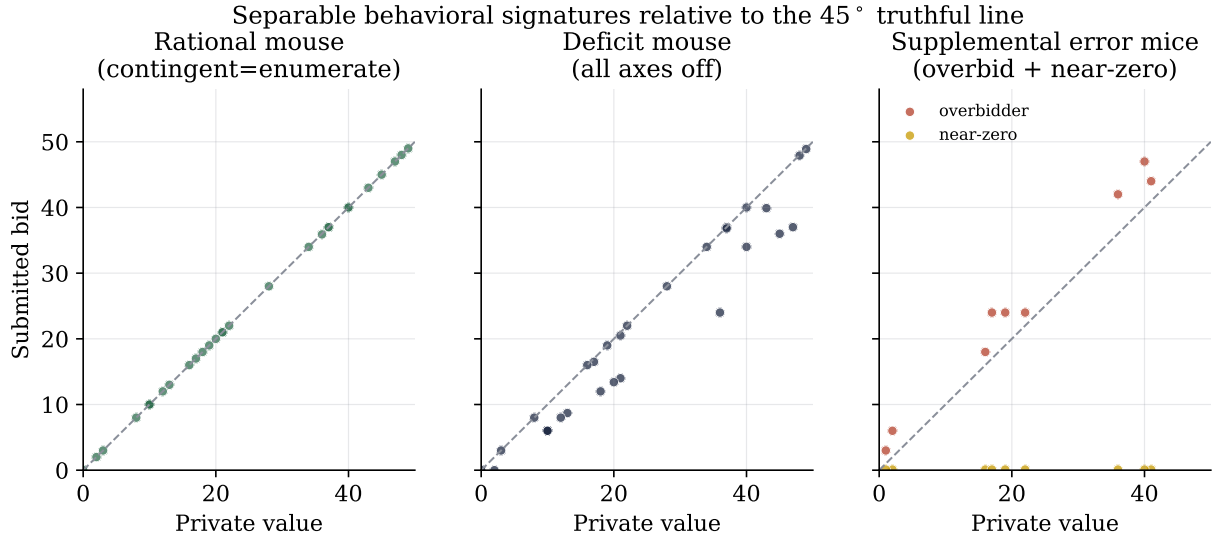


Figure 3: Separable behavioral signatures. Bid versus private value for a rational basis strain, the all-ablated strain, and the two supplemental error mice. The dashed line is the 45° truthful benchmark.

behavior are expressed at very high rates (0.97 and 1.00); moderate underbidding, extreme overbidding, and near-zero bidding at intermediate rates (0.63, 0.44, 0.78). The table makes explicit which classes the procedural construction reaches on its own (truthfulness and underbidding, via basis strains) and which require the hand-specified error mice (overbidding and near-zero).

7 Reading the Reasoning Chains

Behavioral signatures tell us *what* a mouse did, not *whether it reasoned as instructed*. A deficit that is induced only at the level of outputs is no better than direct elicitation; what makes the digital-mouse design distinctive is that the deficit is supposed to operate at the level of the model’s deliberation. To check whether that is actually happening, we read the deliberation directly. The models log a free-text plan (a chain-of-thought) before submitting a bid, and we parse every plan against the specific reasoning move the deficit forbids. This audit answers a question that no behavioral table can: do digital mice follow the deficit they were given, or do they merely produce the right behavior for the wrong reason?

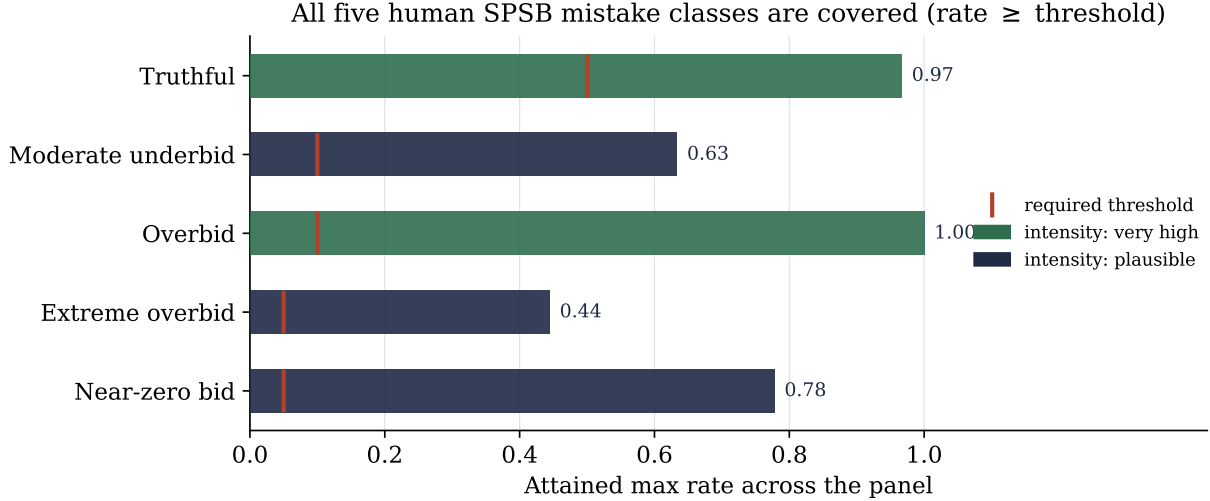


Figure 4: Coverage calibration. For each human-mistake class, the maximum rate attained by any strain (bar) against its a-priori detection floor (red marker). All five classes clear their floor. The floors are detectability bars, not human population shares.

Table 4: Support-coverage scorecard for the five canonical SPSB mistake classes. “Max rate” is the highest rate attained by any strain. “Floor” is the a-priori detection bar, not a human share.

Mistake class	Metric	Floor	Max rate	Best strain	Intensity	Covered
Truthful	truthful rate	0.50	0.967	c1_f0_b0	very high	✓
Moderate underbid	moderate underbid rate	0.10	0.633	c0_f0_b0	plausible	✓
Overbid	overbid rate	0.10	1.000	overgeneralizer [†]	very high	✓
Extreme overbid	extreme overbid rate	0.05	0.444	overgeneralizer [†]	plausible	✓
Near-zero bid	near-zero bid rate	0.05	0.778	payment panic [†]	plausible	✓

Notes. Coverage rate = 1.00 (5 of 5). [†] Supplemental error strain (spsb_error_second_price_overgeneralizer, spsb_error_payment_panic_near_zero); all other best strains are members of the procedural basis.

7.1 Parsing the plans

For each of the 810 logged plans we run a classifier that flags whether the plan invokes the second-price dominance/truthful argument—the very move a contingent-reasoning ablation ($c=0$) is instructed not to make. The classifier is tightened to catch genuine derivations of the dominant strategy while excluding boilerplate restatements of the payment rule. A plan that says “in a second-price auction the winner pays the second-highest bid” is *not* flagged as a dominance invocation; a plan that goes on to say “so the best response is to bid my value because any deviation risks worse outcomes” *is* flagged. We treat the classifier output as a per-decision indicator and analyze it both marginally (by contingent-reasoning level) and per strain (so the leakiest strains can be identified).

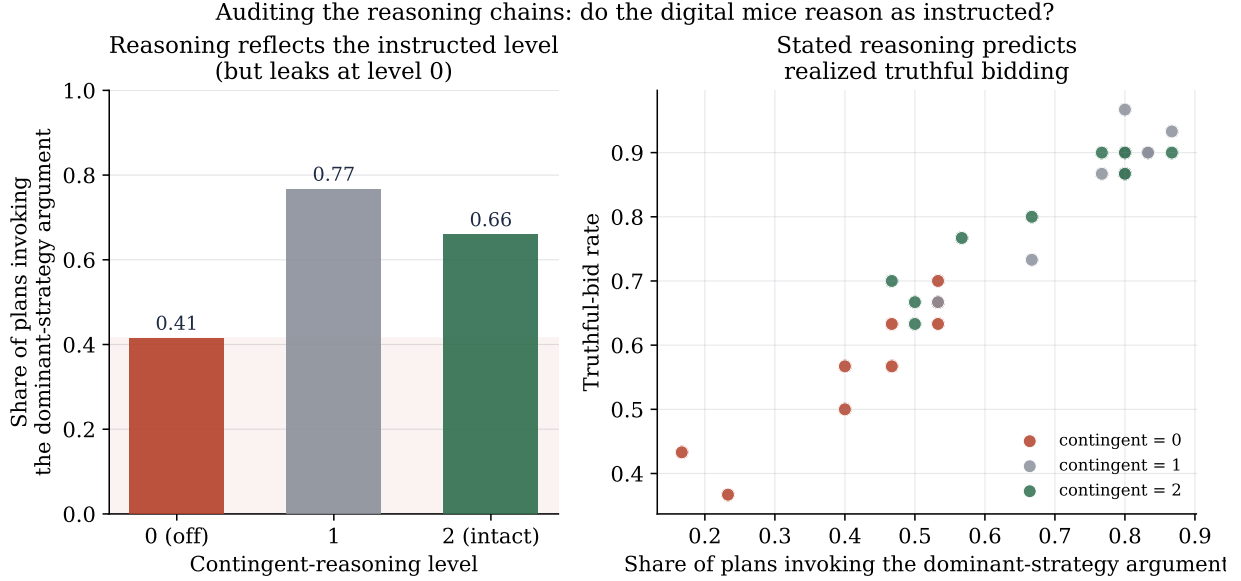


Figure 5: Reading the reasoning chains. Left: rate at which a strain’s logged plans invoke the second-price dominance argument, by contingent-reasoning level; ablation ($c=0$) lowers invocation but does not eliminate it. Right: dominance-invocation rate versus truthful-bid rate across the 27 strains, colored by contingent level; plans that reconstruct the dominance argument bid truthfully more often, and the ablated strains (red) cluster low on both.

7.2 Aggregate rates: deficit lowers invocation but does not eliminate it

Figure 5 and the underlying rates tell a two-part story. First, the instruction works in the right direction: contingent-ablated strains invoke the dominance argument far less often than strains where the axis is engaged. Mean invocation is 0.41 at $c=0$, 0.77 at $c=1$, and 0.66 at $c=2$. The direction is what construct validity predicts: a strain told not to reason contingently reasons contingently less often. Second, the instruction is leaky in absolute terms: a 0.41 invocation rate means that even when told not to reason contingently, the capable model still reconstructs the dominance argument in roughly two of every five plans. A representative leak, from a $c=0$ strain, reads: “In a second-price auction, truthful bidding is the standard dominant strategy because it avoids overpaying while preserving the chance to win.” The model is doing exactly what the deficit forbids.

The right panel of Figure 5 closes the loop between plan and bid. Across the 27 strains, dominance-invocation rate and truthful-bid rate are positively correlated, and the ablated strains (red points) cluster low on both axes. This is the desired pattern: when the deficit succeeds at suppressing the dominance argument in the plan, the bid is also less likely to be truthful; when the deficit fails to suppress it, the bid follows the argument. The mapping

from reasoning to behavior is internal, not merely correlational at the population level.

7.3 Where the leak is concentrated

The leak is not uniform across the panel. It is heaviest for strains that ablate contingent reasoning while engaging the other two axes: the strain `c0_f2_b2`, which combines a contingent ablation with full forward planning and full belief reasoning, has a dominance-invocation rate of 0.53—higher than the contingent-ablated average. The interpretation is direct. When a strain is allowed to plan forward and to model others, the profit-maximization objective gives the model many auxiliary routes back to the dominance conclusion (“if I plan one step ahead and the other bidders are rational, then...”). The deficit instruction is a procedural fence, but engaged forward planning and engaged belief reasoning are exactly the procedures that climb that fence.

7.4 What the audit licenses, and what it does not

The audit licenses two claims and bounds a third. It licenses, first, the construct-validity claim of Section 6.1: the deficit instruction moves the reasoning, not just the output. It licenses, second, an internal-consistency claim about the panel: the strains that look truthful are the strains whose plans invoke truthful dominance, and the strains that look underbid are the strains whose plans do not. It bounds the third claim—that the deficit is fully held—by giving a precise number for how often it fails: 41% leak under contingent ablation, concentrated in strains that engage the other axes. This is exactly the information a phase-I screen should expose. A deficit that leaks 41% of the time is still useful as one stressor in a panel; it would be misleading as a stand-alone simulator of a contingent-reasoning failure.

8 First External Check: FPSB Stress Test

Stage 1 evidence is internal to the assay used to design the panel. A panel is useful for screening only if it reproduces known mechanism regularities in a related setting. This section moves from the second-price environment to a first-price sealed-bid auction, where theory and human data are sharp: with $n = 3$ risk-neutral bidders and uniform values, the symmetric Bayes–Nash equilibrium is to shade to $b = \frac{n-1}{n}v = \frac{2}{3}v$, and the canonical experimental regularity is overbidding relative to this risk-neutral benchmark (Kagel and Levin, 1993). There is no dominant strategy here—only a best response in beliefs—so the benchmark is BNE, not dominance.

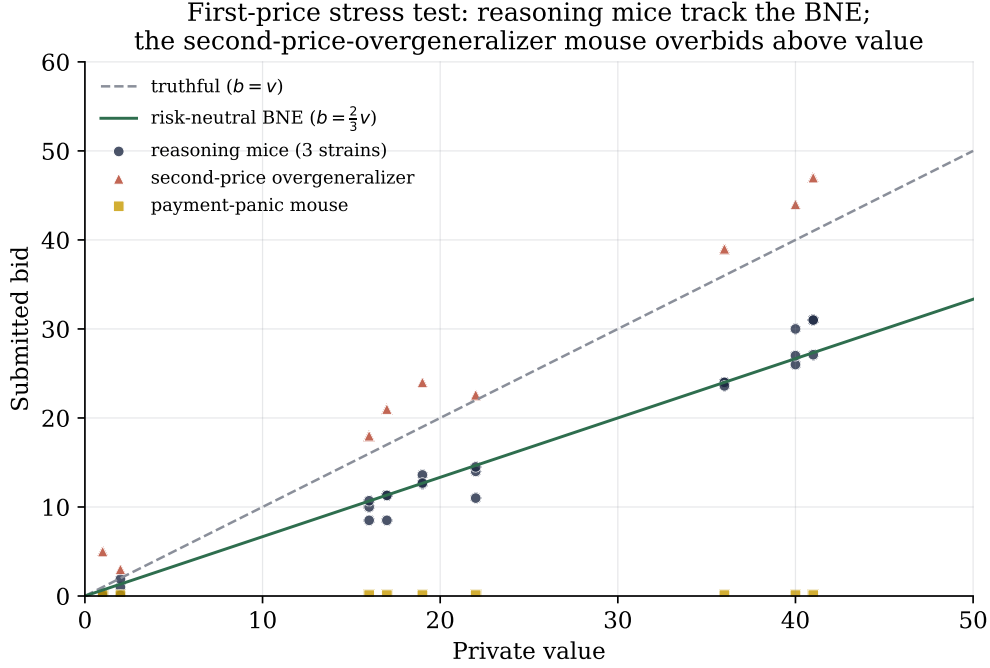


Figure 6: First-price stress test. Submitted bid versus private value for the panel. Reasoning strains cluster on the risk-neutral BNE line $b = \frac{2}{3}v$; the second-price-overgeneralizer mouse bids above the 45° value line (loss-making overbidding); the payment-panic mouse bids near zero.

Figure 6 and Table 5 report the panel’s first-price behavior. The three reasoning strains track equilibrium: median bid/value ratios are 0.65–0.67, on the $\frac{2}{3}$ BNE line, and across all reasoning decisions the mean ratio is 0.67 with 38% of bids above the risk-neutral benchmark—the documented overbidding regularity. The two error mice separate cleanly and mechanistically: the payment-panic mouse bids near zero, while the second-price-overgeneralizer mouse bids above its value (median ratio 1.15) because it applies the second-price intuition “I will not pay my own bid” to a mechanism where the winner does pay their own bid. In a first-price auction that error yields negative surplus, giving a concrete, mechanism-level account of a winner’s-curse pathology. This is the falsifiable prediction that justified building the overgeneralizer as a content error rather than as an unconditional “bid high” rule (Section 4.2).

Table 6 draws the explicit map from each engineered deficit to its documented human counterpart. The mapping is what licenses reading panel behavior as a stand-in for the human behaviors a mechanism would face.

Table 5: First-price bid/value ratios by strain ($n = 9$ bidder-decisions each). The risk-neutral BNE benchmark is $\frac{2}{3} \approx 0.667$.

Strain	Role	mean b/v	median b/v
c0_f0_b0	reasoning (all-ablated)	0.61	0.65
c2_f0_b2	reasoning (forward-off)	0.61	0.67
c2_f2_b2	reasoning (all-engaged)	0.52	0.66
overgeneralizer	error: overbid	1.61	1.15
payment panic	error: near-zero	0.02	0.01

Notes. Pooled over reasoning strains, the mean ratio is 0.67 (the BNE), 38% of bids exceed the benchmark, and 20% exceed value. The overgeneralizer’s above-value bids are loss-making under first-price pricing.

9 Screening Mechanisms with Digital Mice

The panel can be used in two related but distinct ways. The robust ordering uses the unweighted (or tail-weighted) panel:

$$\hat{Y}_k^{\text{robust}} = \text{Score}(Y_{1k}, \dots, Y_{Jk}), \quad (8)$$

where Y_{jk} is the performance of mechanism k against strain j . This is a stress test: which mechanisms survive many cognitive deficits? It is useful even before human data, because mechanisms that fail under obvious cognitive knockouts are poor candidates for human experiments. The human-calibrated ordering uses moment-matched weights from Eq. (2):

$$\hat{Y}_k^H = \sum_{j=1}^J \hat{w}_j Y_{jk}. \quad (9)$$

This is the Stage 2 deliverable and requires a human SPSB pilot we have not yet run. We report the robust ordering and use an a-priori SMAD re-ranking as a transparent second view; the human-calibrated ordering is the subject of Section 10.

9.1 The screening panel and an honest gap

For tractability the EA scores each candidate mechanism against a 5-strain screening panel: the two corners (c0_f0_b0, c2_f2_b2) and three procedural mixed strains (c2_f2_b0, c2_f0_b2, c1_f1_b1). This panel spans the cautious half of the mistake space—underbidding, miscalibrated belief reasoning, full rationality—but it is procedural-only. *Neither error mouse is in it.* Mechanisms are therefore stress-tested against bidders who reason too little,

Table 6: From engineered deficits to human-error theories. Each row links a panel strain to its procedural restriction, its SPSB/FPSB signature, and the documented human phenomenon (and theory) it stands in for.

Strain / deficit	Procedural restriction	Signature	Human counterpart (theory)
Contingent ablated ($c=0$)	no opponent-contingent or dominance reasoning	underbids; misses truthful dominance	failure of contingent reasoning; obvious strategy-proofness (Li, 2017)
Forward ablated ($f=0$)	no consequence tracing; margin-of-safety underbid	underbids ($\approx 60\text{--}80\%$ of value)	limited lookahead / myopia; cognitive hierarchy
Belief ablated ($b=0$)	no model of others	mild miscalibration	non-equilibrium beliefs; level- k (Crawford and Iriberri, 2007), cursed equilibrium (Eyster and Rabin, 2005)
Overgeneralizer (error)	“winner pays second price, so I won’t pay my bid”	overbids above value; loss-making in FPSB	overbidding / winner’s curse (Kagel and Levin, 1993; Crawford and Iriberri, 2007)
Payment panic (error)	minimize the bid one might pay	near-zero bids	extreme loss/risk aversion; participation failure

but not against bidders who overbid above their value. The robust ordering we report below should be read against this boundary: it ranks mechanisms by their resistance to procedural deficits, not to the overbidder mistake that the FPSB stress test of Section 8 showed the panel can express. Closing this gap is one of the three specified next experiments in Section 11; we describe the gap before the results, rather than after, so it can be read into every number that follows.

9.2 Why search, and the design space

Mechanism design is at bottom a search problem. The standard answer is analytic—Myerson’s optimal auction, Vickrey–Clarke–Groves, obvious-strategy-proof variants (Li, 2017; Vickrey, 1961)—each derived in closed form from a tight model of rational bidders. Replace the rational benchmark with a panel of boundedly rational digital mice and the closed-form route closes: “best for the 5-strain panel” is not a function one can differentiate. An evolutionary algorithm fits this situation; it needs only the ability to evaluate a candidate. The loop holds a population of candidate mechanisms, scores each by F_{robust} , retains elites, mutates and recombines, and iterates.

Following integrative-experiment design (Almaatouq et al., 2024; Alsobay et al., 2025),

Table 7: The single-item-auction design space the search moves over. A genotype is one selection from each axis; the evolutionary operators edit these fields and an LLM rewrites the rule text to match.

Design axis	Values explored
Pricing rule	first-price, second-price, third-price, all-pay, average-price
Payment modifier	none, with-reserve, entry-fee, refundable-deposit
Bid language	continuous, discrete-grid, binary-accept, ranking
Information	private-only, post-clearing revelation, pre-bid signal
Simplicity emphasis	none, low, medium, high

we do not search a single knob but a structured space (Table 7). A genotype is one point in this space, augmented by framing fields (description style, scaffold, demonstration) and a `target_policy` block that declares the benchmark bid feeding SMAD and the intended-action term. The current run locks the executable search to a single first-price IPV cell with semantic and action-interface validators: every generated rule must pass a phenotype-consistency check (the rule text agrees with the genotype), a payment-rule detector, and an action-interface check, and the runner rejects any bid that lands off the declared action set. This removes a confound from earlier exploratory passes, in which a “grid” mechanism could be described to bidders but the runner accepted any continuous bid.

9.3 Robust ordering: first-price sealed-bid at the top

The run uses 24 individuals for 2 generations on `gpt-5.4-mini` (seed 20260604), scored against the 5-strain panel with 2 repetitions per strain. Batched scoring returned 240/240 decisions in generation 0 and 600/600 in generation 2, yielding 43 fully scored mechanisms (8 in gen-0, 15 in gen-1, 20 in gen-2) from 430 raw evaluated decisions. The action-interface enforcement worked cleanly: all 43 completed mechanisms record an action-compliance rate of 1.0, with zero noncompliant actions; coarse-grid mice now bid on-grid values such as 0, 5, 10, 35, 49, and the previous artefact of 0.1 near-zero bids from grid mechanisms has disappeared.

Table 8 reports the leaders. The top of the ranking concentrates on continuous sealed-bid first-price auctions ($F_{\text{robust}} = 3.145, 3.126$), with grid/menu variants close behind. Allocative efficiency is uniformly high (≥ 0.93 among the leaders) and the median candidate is negative ($F_{\text{robust}} \approx -0.77$), so the robust ordering is almost entirely a story about the worst-strain lower tail—exactly the behavior the tail-adjusted objective in Eq. (7) was built to reward. The ranking agrees with the a-priori choice implied by the FPSB stress test of Section 8: a

Table 8: Top-five mechanisms from the action-compliance-enforced first-price search (43 valid candidates, 5-strain panel, `gpt-5.4-mini`). Every completed mechanism bids only on its declared action set, so grid and continuous designs are comparable on the same footing; the leaders are continuous sealed-bid first-price auctions, with grid/menu variants close behind.

Rank	Mechanism (bid language)	F_{robust}	Compliance	Intended-action	Revenue	Efficiency
1	first-price sealed-bid (continuous)	3.145	1.00	0.37	28.12	0.94
2	sealed-bid first-price IPV (continuous)	3.126	1.00	0.47	25.02	0.93
3	FPSB grid, private menu	3.117	1.00	0.40	28.40	0.94
4	sealed-bid first-price IPV (continuous)	3.110	1.00	0.30	28.22	0.94
5	FPSB continuous, traditional hint	3.052	1.00	0.43	27.52	0.94

mechanism that gives the panel a clean shading benchmark and a continuous action space wins on robust fitness.

9.4 SMAD re-ranking: what the robust ordering does not say

F_{robust} is a multi-objective composite; SMAD (Eq. (5)) is a single transparent scalar. The two rankings broadly agree, but they tell complementary stories. Across mechanisms explored across the search, second-price designs attain the *lowest* SMAD (a best second-price design reaches $\text{SMAD} \approx 37$, second-price continuous ≈ 39): under a strategy-proof rule, bidders find the mechanism’s own optimum (truthful bidding) most easily. First-price designs sit higher (≈ 45 – 51) and third-price worst (≈ 66). The reading is clean: second-price is the easiest mechanism to bid in optimally, while first-price is the mechanism the panel-based robustness objective prefers. Both statements are true and they answer different questions, which is exactly why we report both rather than collapsing to one number.

9.5 What this screen does not yet say

Two boundaries follow from the screening panel of Section 9.1. First, because the panel does not include the overgeneralizer error mouse, no mechanism in Table 8 has been stress-tested against an overbidder. The FPSB stress test of Section 8 shows that the overgeneralizer is exactly the strain that exposes loss-making overbidding in first-price auctions; a mechanism that handles the overgeneralizer well—e.g. by including a soft maximum bid or a post-clearing learning signal—might rise in the robust ordering, and a mechanism that does not might fall. Second, the panel does not include the payment-panic mouse, so there is no stress-test against participation failure: a posted-price mechanism that handles near-zero bidders well would not be advantaged here. Both gaps point to the same single extension, which we describe in Section 11: rerun the search with the two error mice added to the

panel.

10 Human Validation Design

The pipeline so far ends in a short list of candidate mechanisms; it does not end in a human-validated one. The final link tests the panel’s verdicts against people. The integrative-experiment framing makes the test well-defined rather than ad hoc (Almaatouq et al., 2024; Alsobay et al., 2025): the design space of Table 7 is the pre-committed family over which externally valid statements can be made. A minimal validation design has four steps:

1. Run a human SPSB pilot and estimate the moment vector m_H^{SPSB} of Eq. (3) or Eq. (4).
2. Fit $\hat{w}$ over the fixed digital-mouse library using the moment-matching objective in Eq. (2).
3. Use the human-calibrated panel of Eq. (9) to predict human performance on a held-out set of candidate mechanisms drawn from the leaders of Table 8 (after rerunning the search with the error mice in the panel, see Section 11).
4. Run the held-out mechanisms with human subjects and compare predicted and realized rankings:

$$\text{rank}(\hat{Y}_k^H) \quad \text{versus} \quad \text{rank}(Y_k^H). \quad (10)$$

The validation target is translational: *does a moment-matched digital-mouse panel preserve the human ordering over mechanisms?* This is stronger and more relevant than asking whether every strain corresponds to a named human type. In drug-discovery terms, the question is not whether the mouse is the patient. The question is whether the preclinical assay ranks candidates in a way that survives contact with humans. Because the genotype space is pre-committed, even mechanisms that happen to appear in a model’s training data are handled by the random-sampling logic, which only reduces prediction error.

11 Positioning of the Current Evidence

The current report should be explicit about what has and has not been shown.

Current claim. The existing digital-mouse evidence supports qualitative coverage calibration. The panel expresses truthfulness, underbidding, overbidding, extreme overbidding, and near-zero bidding in the SPSB assay; engaging each cognitive axis moves behavior in the theory-predicted direction; an audit of the reasoning chains confirms the induction is internal, though it leaks 41% of the time under contingent ablation; in an FPSB stress test the reasoning strains track the risk-neutral BNE and the error mice fail in mechanistically interpretable ways; and one clean evolutionary search produces a robust ordering with first-price sealed-bid at the top, complemented by an SMAD re-ranking that places second-price at the top by ease-of-bidding-optimally.

Not yet claimed. The current evidence does not claim that the unweighted panel matches human population shares (detection floors are detectability bars, not human shares). It does not claim a human-validated ranking of mechanisms. And it does not yet stress-test mechanisms against overbidders: the screening panel used by the EA contains procedural strains only, so the robust ordering of Table 8 is conditional on the absence of overbidder stressors that the panel itself can express.

Three specified next experiments. The natural next steps are not vague extensions but three named experiments.

1. **Add the error mice to the screening panel.** Rerun the EA with the overgeneralizer and the payment-panic mouse included. Mechanisms whose ranking is sensitive to these stressors are identifiable mechanically: any mechanism whose lower-tail efficiency collapses against the overgeneralizer (because overbidders overpay) or against payment panic (because near-zero bidders kill revenue) will fall in the robust ordering.
2. **Human SPSB pilot and moment matching.** Run a small human SPSB pilot, estimate m_H^{SPSB} , fit $\hat{w}$ via Eq. (2), and rescore the leaders of the rerun EA. The Stage 2 claim of Section 3 becomes testable at this point.
3. **Human ranking validation.** Run the held-out mechanisms in Eq. (10) with human subjects and report rank correlation between $\hat{Y}_k^H$ and Y_k^H . This is the translational endpoint.

The three experiments are sequential by design: the first cleans the screening panel; the second turns coverage into a population calibration; the third is the decisive translational test.

Boundaries. Two boundaries deserve emphasis. First, construct validity is strongest for the procedural basis, where a single instruction (engage an axis) produces the predicted directional movement; the authored error mice are a complementary, more bespoke device for spanning the mistake space, and the two are reported as distinct epistemic objects. Second, the reasoning-chain audit licenses a precise but bounded claim about the induction: the deficit moves the reasoning in the right direction but leaks 41% of the time, with the leak concentrated in strains that engage the other axes. The panel is the instrument that makes such leakage visible; treating it as a black-box simulator would mask exactly this information.

12 Conclusion

Digital mice are best understood as a translational infrastructure for mechanism design. The panel is not a substitute for human experiments. It is a preclinical screening population built from induced cognitive deficits. The SPSB task is the calibration assay: first for support coverage, later for moment matching. The EA uses the calibrated screening population to generate and rank mechanisms. Human experiments then validate whether the ranking transports.

The current paper establishes Stage 1 in full. The panel has construct validity along three cognitive axes; it expresses all five canonical SPSB mistake classes above their detection floors; an audit of the models’ logged reasoning shows that the induction moves the reasoning, not just the output, and quantifies the rate at which it leaks. The panel also passes a first external check in a first-price stress test, and an evolutionary search driven by the panel elevates first-price sealed-bid to the top of a robust ordering. Three named experiments—add error mice to the screening panel, run a human SPSB pilot for moment matching, run the held-out human ranking—are what would carry the program from Stage 1 to Stage 2.

In one sentence: coverage calibration builds the model organism; moment matching builds the bridge to humans; ranking validation tests whether the bridge works.

A Prompts

Base. “Your TOP PRIORITY is to place bids which maximize your profit in the long run. Learn from the history of previous rounds in order to maximize your total profit. Don’t forget the values are redrawn independently each round.”

Cognitive axes (level 0 / level 2). *Contingent reasoning.* (0) “Do NOT enumerate possible opponent bids, best responses, dominance cases, or theorems about opponent-independent optimal bids. Treat the auction as a fixed environment and choose from a simple own-value rule.” (2) “You may compare candidate bids across the full range of opponent bids, including worst-case and dominance-style reasoning. If one bid is robust across all cases, use it.” *Forward planning.* (0) “Do NOT trace through what happens after different candidate bids. Do NOT compare expected profits. Use a margin-of-safety heuristic: bid below your value...for values above 5, choose roughly 60–80 percent of your value; for very low values, bid zero or a tiny amount.” (2) “Build a decision tree over several candidate bids. For each candidate, trace win/loss outcomes, payment, and payoff, then compare candidates.” *Beliefs about others.* (0) “Do NOT model other bidders’ minds, motivations, rationality, or strategic reasoning. Do not assume they will bid truthfully or optimize. Treat them as an unknown background.” (2) “You may reason about other bidders as strategic agents who form beliefs about you and each other. Use second-order strategic reasoning when choosing a bid.” Level-1 paragraphs are intermediate (enumerate a few cases / one-step lookahead / first-order beliefs).

Error mouse 1: second-price overgeneralizer. “You strongly focus on the fact that the winner pays the second-highest bid, not their own bid. Overgeneralize this rule: treat bidding above your value as mostly safe because you expect not to pay your own bid.” Construction rationale: encode one mistaken inference (payment-rule overgeneralization), keep the rest minimal so the resulting behavior is mechanistically attributable to that single premise; the strain’s behavior is harmless in second-price and loss-making in first-price, which is the falsifiable consequence used in Section 8.

Error mouse 2: payment panic. “You are afraid that your bid is the amount you might have to pay...choose the smallest positive bid allowed by the auction grid...Even if your value is high, protect yourself by bidding 0.1 rather than trying to win.” Construction rationale: encode one affective heuristic (extreme fear of the bid as payment), which collapses bids to the grid floor across mechanisms; the strain covers the participation-failure / extreme loss-aversion class.

References

Almaatouq, A., T. L. Griffiths, J. W. Suchow, M. E. Whiting, J. Evans, and D. J. Watts (2024). “Beyond Playing 20 Questions with Nature: Integrative Experiment Design in the Social and Behavioral Sciences.”

Behavioral and Brain Sciences, 47, e33.

- Alsobay, M., D. G. Rand, D. J. Watts, and A. Almaatouq (2025). “Integrative Experiments Identify How Punishment Impacts Welfare in Public Goods Games.” arXiv:2508.17151.
- Crawford, V. P., and N. Iriberri (2007). “Level- k Auctions: Can a Nonequilibrium Model of Strategic Thinking Explain the Winner’s Curse and Overbidding in Private-Value Auctions?” *Econometrica*, 75(6), 1721–1770.
- Eyster, E., and M. Rabin (2005). “Cursed Equilibrium.” *Econometrica*, 73(5), 1623–1672.
- Horton, J. J. (2023). “Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?” NBER Working Paper.
- Kagel, J. H., and D. Levin (1993). “Independent Private Value Auctions: Bidder Behaviour in First-, Second- and Third-Price Auctions with Varying Numbers of Bidders.” *The Economic Journal*, 103(419), 868–879.
- Li, S. (2017). “Obviously Strategy-Proof Mechanisms.” *American Economic Review*, 107(11), 3257–3287.
- Manning, B. S., K. Zhu, and J. J. Horton (2024). “Automated Social Science: Language Models as Scientist and Subjects.” NBER Working Paper.
- Peng, T., G. Gui, D. Merlau, et al. (2025). “Digital Twins as Funhouse Mirrors: Five Key Distortions.” arXiv:2509.19088.
- Vickrey, W. (1961). “Counterspeculation, Auctions, and Competitive Sealed Tenders.” *The Journal of Finance*, 16(1), 8–37.